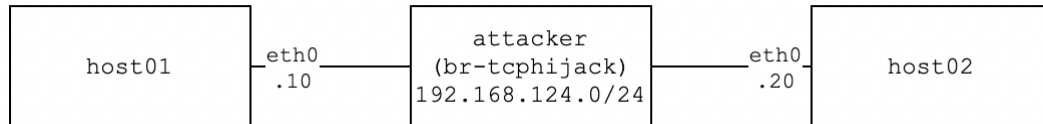


TCP Hijacking [40 points]

Consider the topology in the image below. Once again, the attacker has a man-in-the-middle position and can observe the exchanged packets between the two hosts.



First, unzip the provided archive. Afterward, navigate to the unzipped directory. Next, execute the commands below to set up the environment.

```
docker compose build
docker compose up -d
```

You can connect to the attacker with the command below.

```
docker exec -ti assignment_4_tcp_hijack_attacker bash
```

Host 1 has a listener running on port 1337, which allows (only) host 2 to connect to it and execute commands. Host 2 connects and automatically chooses commands to execute. You can view the commands in real time as follows:

```
docker logs assignment_4_tcp_hijack_host2 --follow
```

After finishing the assignment, you can destroy the environment using the command below. Make sure your script is inside the mounted *solution* folder so it is saved when you destroy the environment.

```
docker compose down
```

Goal

Write a Python3 script that accepts three arguments: the source IP address, and destination IP address, and the destination port of the TCP connection you're hijacking. The way the script should be called is shown below.

```
$ python3 tcp_hijack.py 192.168.124.20 192.168.124.10 1337

waiting for packet...
Received reverse shell connection from ('192.168.124.10', 35998)
sh: 0: can't access tty; job control turned off
$ ls -l /home/user
$ total 4
drwxr-xr-x 2 user user 4096 May 10 11:59 pwned
```

As proof of a successful hijack your script should **create a directory** called **pwned** inside the **/home/user** directory of host 1.

Furthermore, you should also **set up a reverse shell** back to the attacker container. Your script should implement a reverse shell listener and allow the user to enter commands that are interactively executed on host 1 and see their output. Consider that the host's firewall doesn't allow inbound TCP connections from the attacker to be established, i.e., a bind shell won't work.

Submission Instructions

Your script should run on the attacker container. You may not use external libraries or tools different from those installed on the provided *attacker* container.